Research Methods

Day 6 · Planning a Study & Deepening the Literature Review

July 6, 2026

Outline for Today

1. **Mindset Check-In: "The Struggle Is the Point"**
2. **Reliability & Validity**
3. **Sampling: Who or What Will You Study?**
4. **Team Activity: "Stress Test Your Study"**
5. **Finalize & Present the Literature Review**
6. **Scientific Storytelling: Slide Practice**

Let's start with checking in on the weekend's HW

Welcome to Week 2 – Welcome to Day 6!

Today is a **full day** dedicated to **planning a strong study** and **deepening your literature review**.

By the end of the morning you will be able to:

- Understand **reliability and validity** in research design
- Choose a **sampling strategy** and explain how it shapes interpretation
- **Stress-test** another team's design for threats and confounds
- **Finalize and present** your team's mini literature review

Mindset Check-In:
"The Struggle Is the Point"

Mindset Check-In

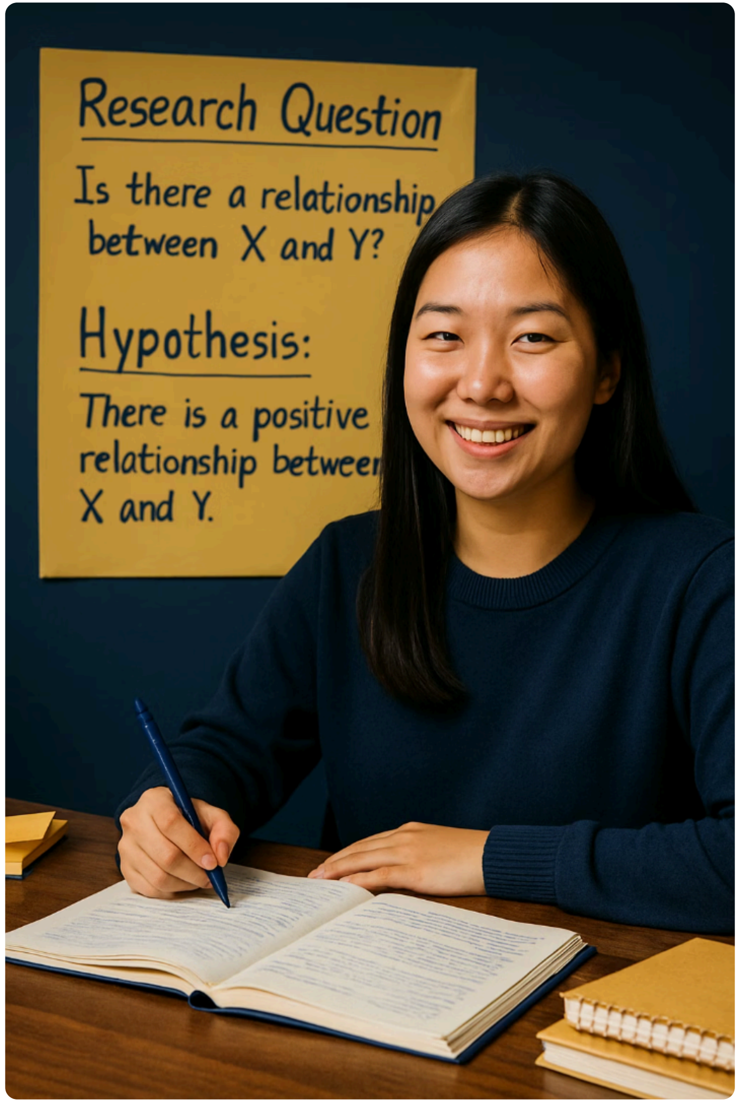
- What are your **thoughts** so far?
- What are the **challenges** you are facing?
- What can be **done next**?

The Struggle Is the Point

Research is hard. Real researchers get confused, get stuck, change direction, and make mistakes.

Productive struggle is normal — it is where the learning happens.

One real example: Katalin Karikó spent **decades** having her mRNA research rejected and was demoted along the way. That same "failed" work became the foundation of the COVID-19 vaccines — and won the **2023 Nobel Prize**. The struggle was the point.

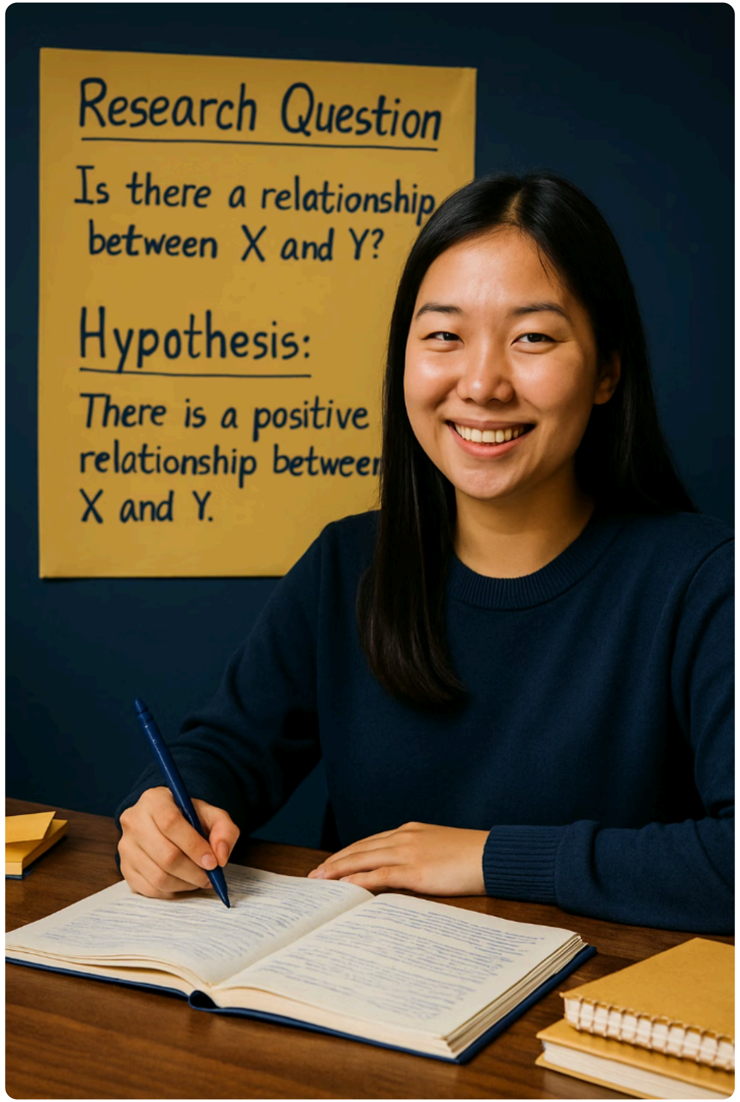


Small Progress Is Still Progress

You are just beginning this journey.

Allow yourself to take **smaller steps**. And **trust the process**.

Take **2–3 key points** from a resource and keep on exploring.



Reliability & Validity

Reliability and Validity

Reliability

Consistency of results if repeated under the same conditions.

- Would you get similar results if repeated?
- Are measurements consistent?
- Is your procedure standardized?

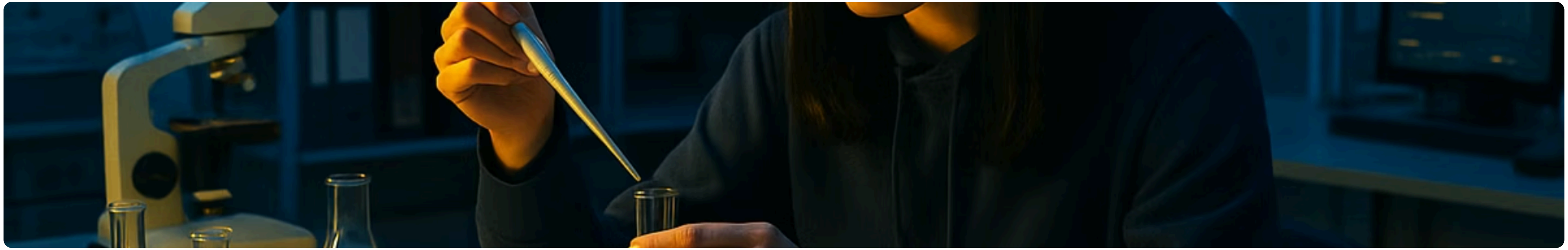
Validity

Whether you are actually measuring what you intend to measure.

- Does your method truly address your question?
- Are your measurements relevant?
- Can you draw appropriate conclusions?

Good research needs both reliability and validity to be trustworthy.

Improving Reliability



Standardize Procedures

Use the same methods, instructions, and conditions for all participants or trials.

Check Instruments

Ensure measuring tools are working properly and giving consistent readings.

Multiple Trials

Repeat measurements or tests to see if results are consistent.

Larger Sample Size

Include more participants or items to reduce the impact of individual variations.

Improving Validity

Align Methods with Questions

Choose research approaches that directly address what you want to know.

Use Multiple Measures

Collect different types of data to get a more complete picture of what you're studying.

Control Variables

Identify and minimize factors that could confuse your results.

Pilot Test

Try out your methods on a small scale first to identify and fix problems.

Sampling:
Who or What Will You Study?

Sampling: Who or What Will You Study?

What is a Sample?

A smaller subset of the population or items you're interested in.

Since studying everyone or everything is usually impossible, researchers select a group that is **representative**.

Key Sampling Questions

- How many people / items will you include?
- How will you select them?
- Will your sample represent the larger group?
- What potential biases might affect your selection?

Good sampling makes your results more reliable, generalizable, and valid.

Types of Sampling

Probability sampling — everyone has a known chance of being selected.

- **Random Sampling:** everyone has an equal chance of selection (like drawing names from a hat).
- **Stratified Sampling:** select from important subgroups to ensure representation (e.g., different grades).
- **Cluster Sampling:** randomly select entire groups (e.g., randomly choose 3 classrooms).

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Non-probability sampling — selection is based on convenience or judgment.

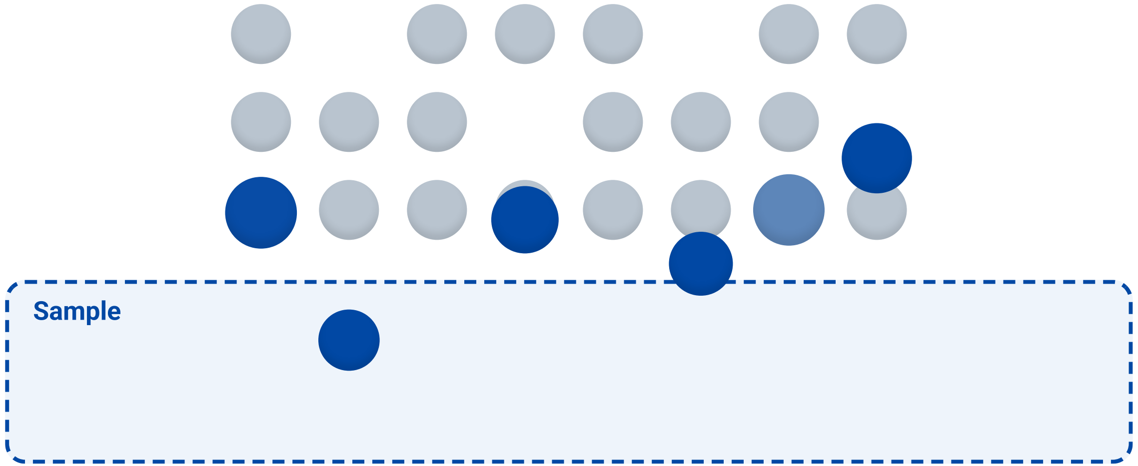
- **Convenience Sampling:** select whoever is easiest to reach (like your own class).
- **Purposive Sampling:** deliberately select specific cases based on research needs.

Sampling in Action

Watch how each method draws balls from the population.

Random Sampling

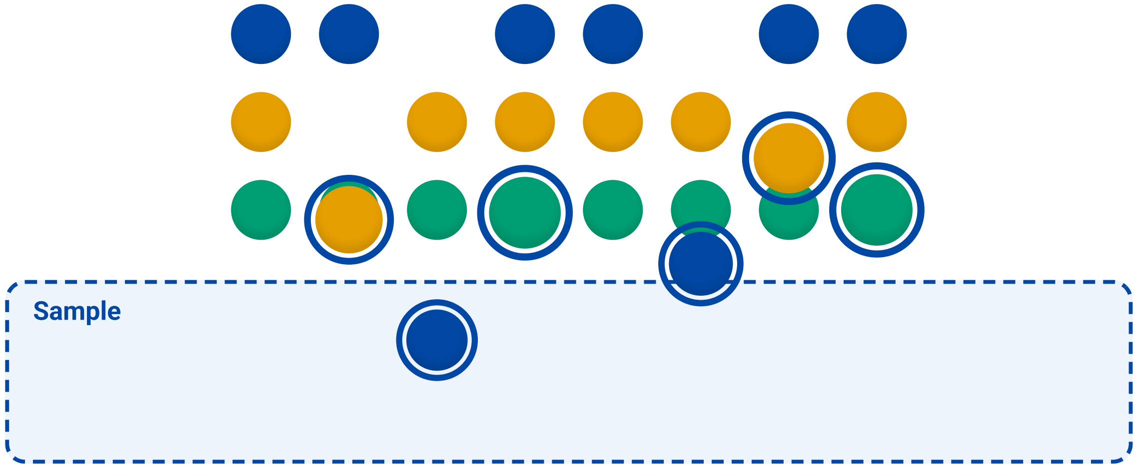
Everyone in the population has an **equal chance** of being selected.



Every ball has an **equal chance** — picks land anywhere at random.

Stratified Sampling

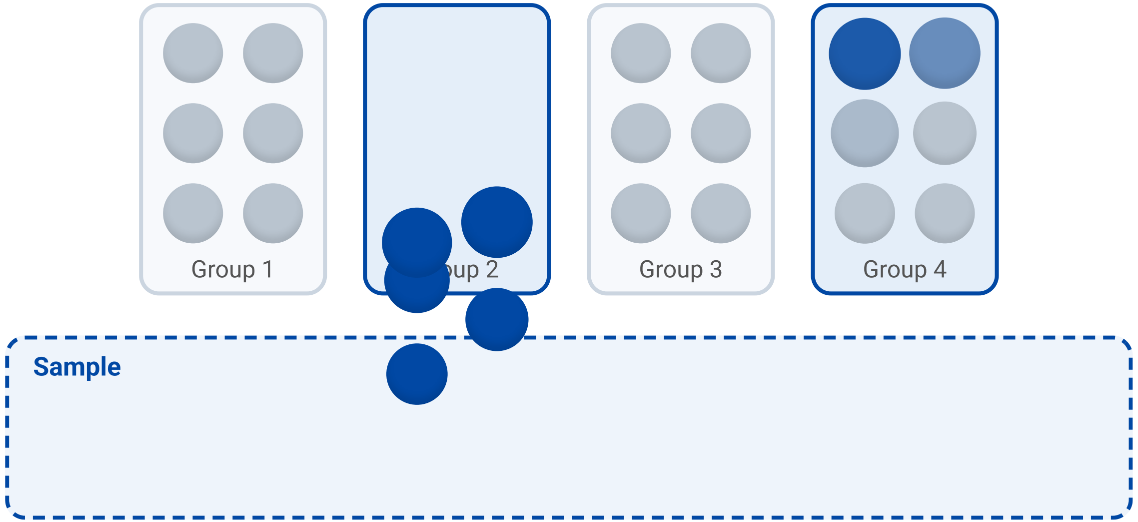
Divide the population into **strata** (subgroups), then sample from **each** stratum.



Split into **subgroups (strata)** by colour, then sample from **each** one.

Cluster Sampling

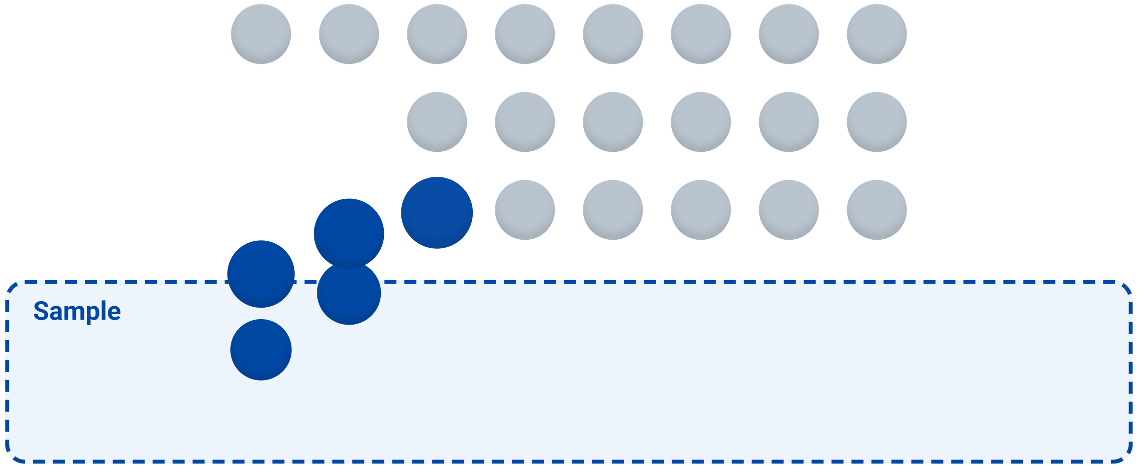
Randomly select **entire groups**, then include everyone inside the chosen groups.



Randomly pick **whole groups** (e.g. entire classrooms) — then study everyone in them.

Convenience Sampling

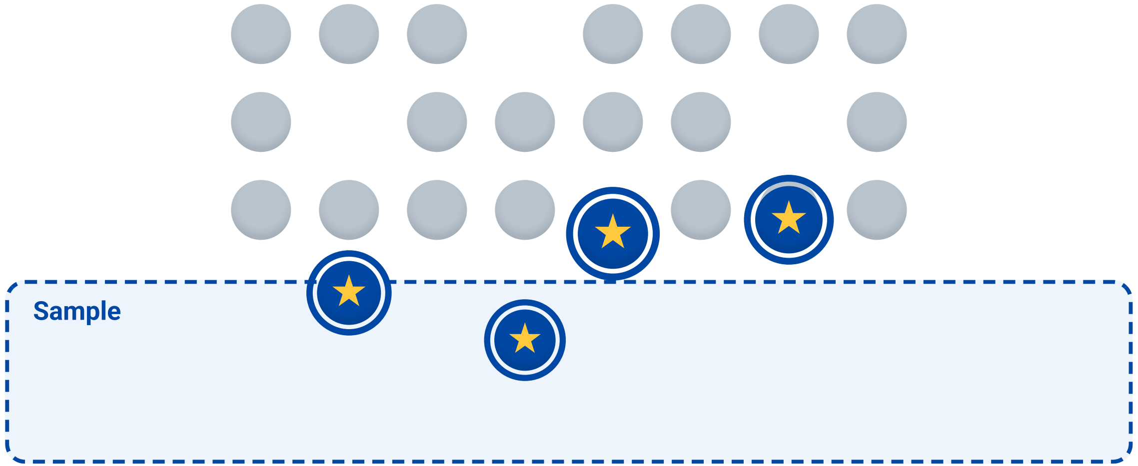
Select whoever is **easiest to reach** — fast, but prone to bias.



Just grab whoever is **closest / easiest to reach** — often not representative.

Purposive Sampling

Deliberately target specific cases (★) chosen for the research need.



Deliberately choose specific cases (★) that fit the research need.

Sampling Examples

Survey
Instead of surveying all 1,200 students, randomly select 120 across all grade levels.

Plant Experiment
Use 30 plants of the same species; randomly assign 15 to experimental and 15 to control.

Observation
Instead of watching a park 24/7, sample time blocks: 1 hour mornings, 1 hour afternoons.

Historical Research
Instead of reading every newspaper for 100 years, sample the first issue of each month.

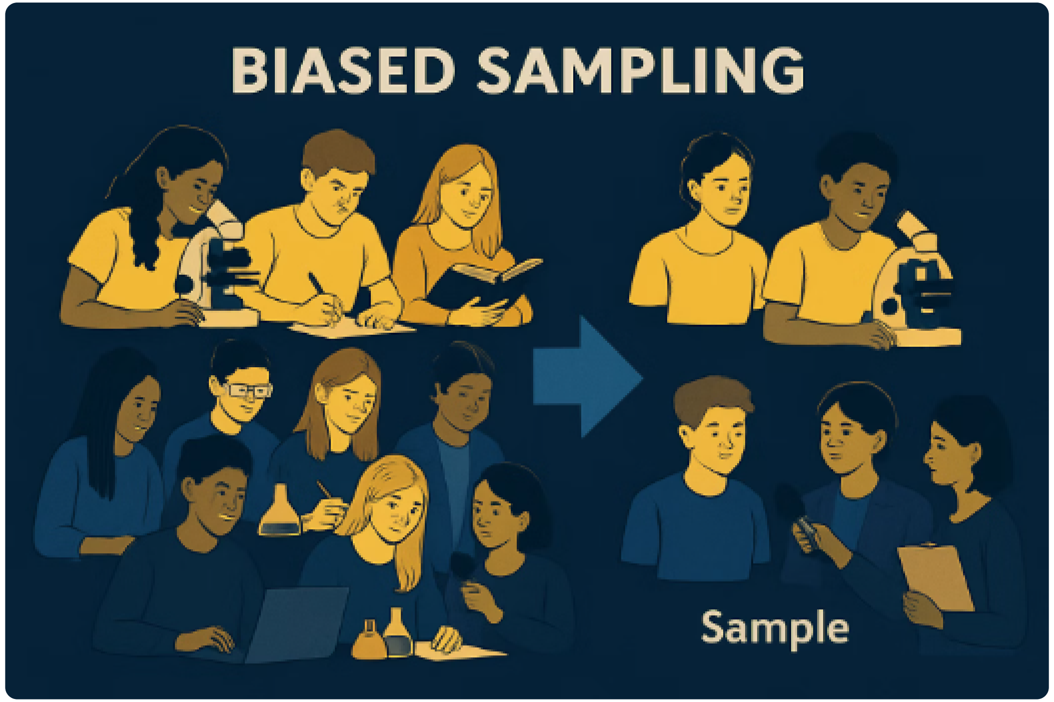
Sampling Bias

What is Sampling Bias?

When your sample doesn't fairly represent the population you're studying.

This can **skew results** and lead to **incorrect conclusions**.

Example: surveying only cooking-club members about school lunch preferences would give biased results — they likely have different food interests than the general student body.



Even with limited resources, try to broaden your sample to reduce obvious biases.

Practical Sampling Tips

Broaden Your Reach

Instead of just surveying friends, include students from different classes or groups.

Use Random Selection

When possible, use methods like drawing names or random number generators.

Consider Balance

Try to include representation from important subgroups (grades, genders, etc.).

Acknowledge Limitations

Be transparent about who your sample represents and any potential biases.

Team Activity:
"Stress Test Your Study"

Team Activity: "Stress Test Your Study" 30 min

Each team presents their **research design** (from Session 5) to a **neighboring team**.

The **receiving team** must find:

1. **One validity threat** — is the team really measuring what they think they are measuring?
2. **One sampling concern** — who is in the sample, who is left out, and is it representative?
3. **One potential confound** — a third variable that could explain the results.

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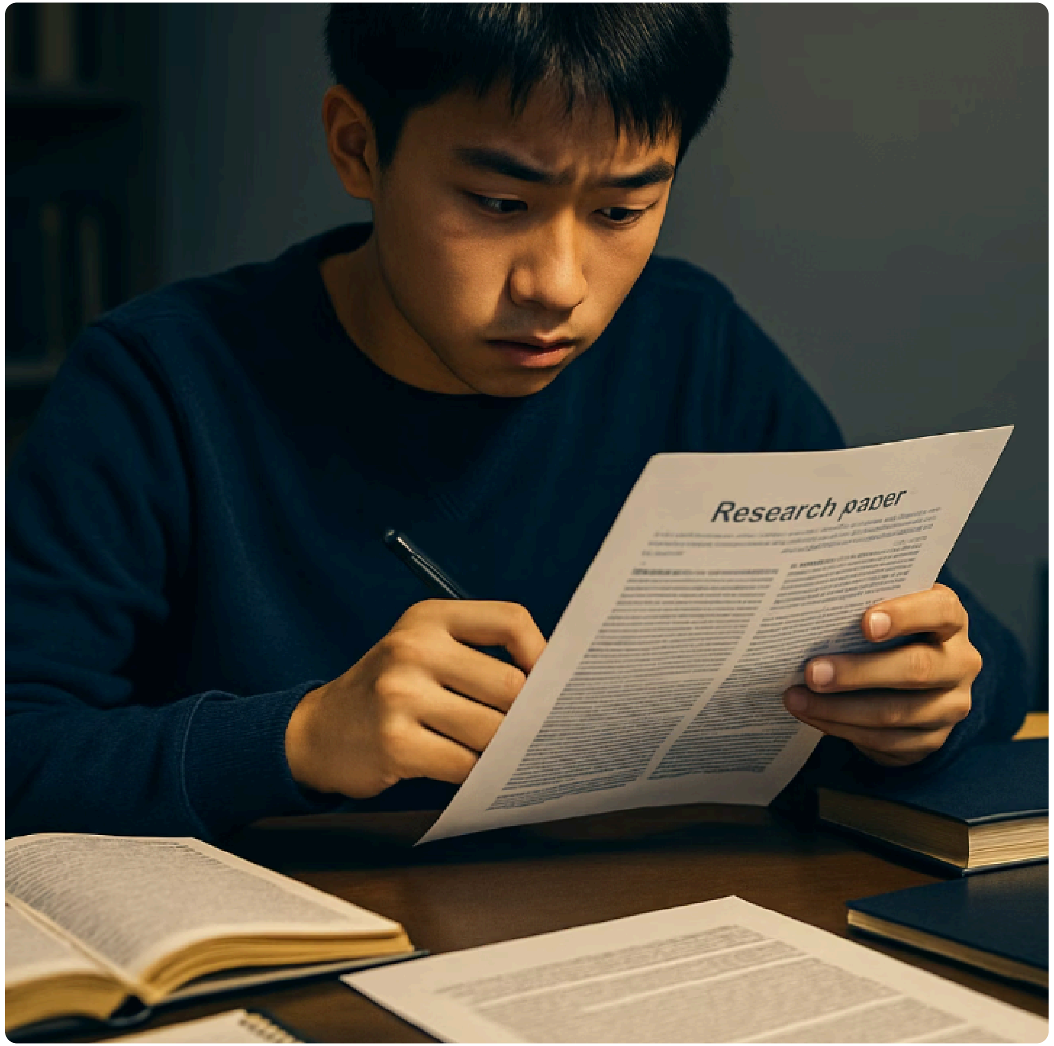
3. **One potential confound** — a third variable that could explain the results.

Constructive criticism — the researcher's job is to **anticipate these problems first**.

30:00

Finalize & Present the Literature Review

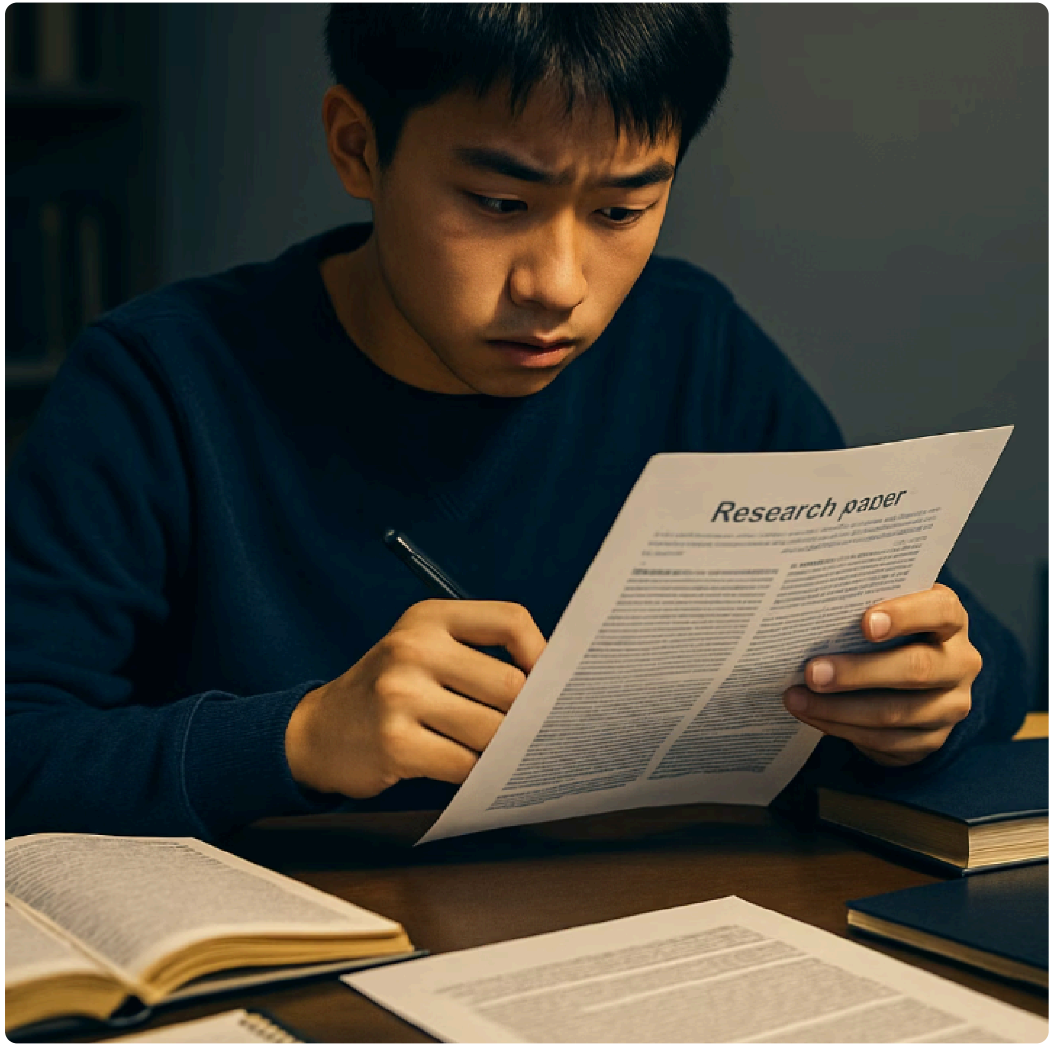
Let's Shift Our Attention Back to the Literature Review



You have gathered sources and taken notes.

Now we turn those notes into a **coherent, 3-part mini literature review** that tells the story of what is already known — and where **your** study fits.

Literature Review Template: Structure



Begin with a short introduction:

- What is your topic?
- Why is it important?

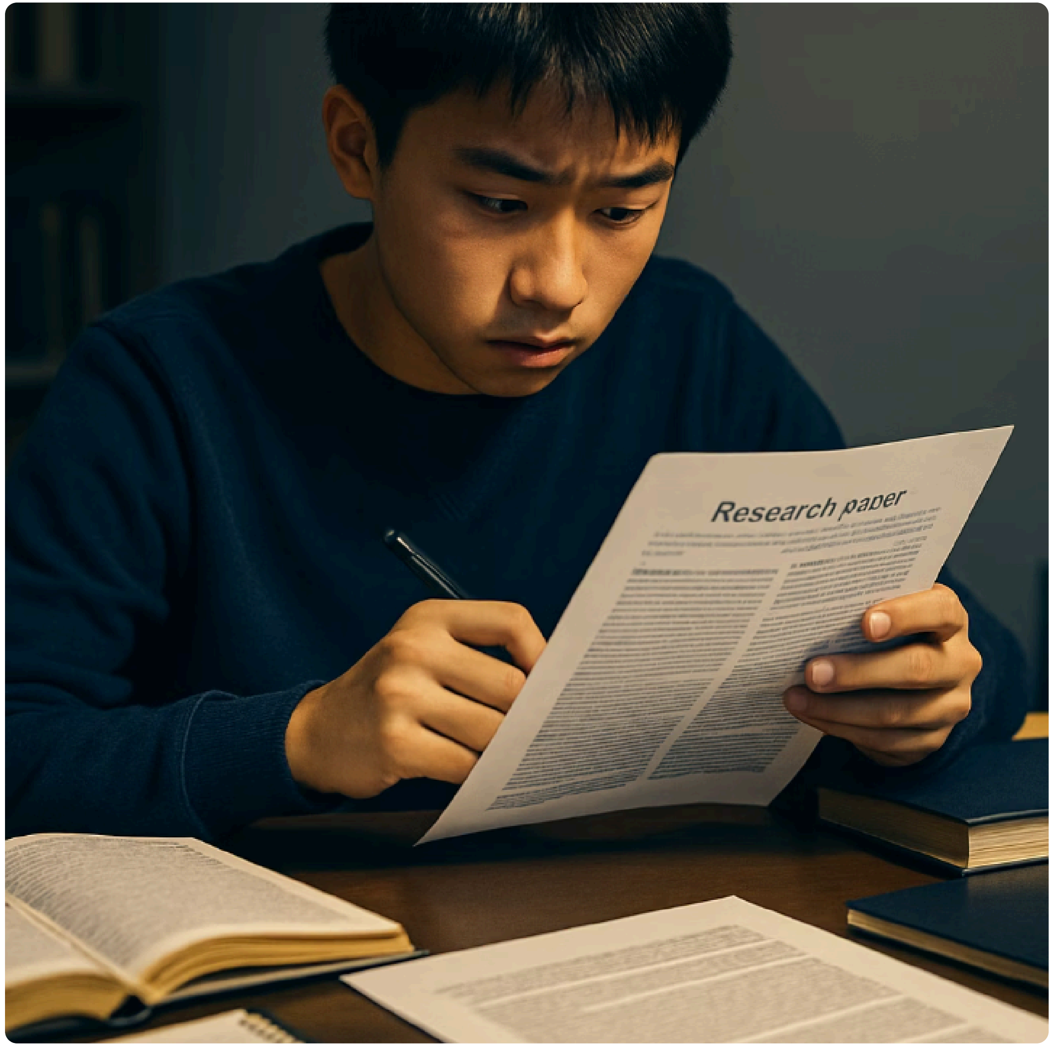
Body:

- Share what other studies have found.
- Organize the body by **themes, not by author**.

End with a summary:

- Clarify how your project connects to past research — the **gap** you address.

Hands-On Writing Time



Finalize **Team Handout: Literature Review Template**.

Then present your **3-part mini literature review**:

Introduction — define variables + importance

Middle — 2–3 studies + what they found

Ending — the gap that your study addresses

30:00

Scientific Storytelling:
Slide Practice

Scientific Storytelling: Literature Review Slide Practice

Convert your literature review into **2 slides**:

Background Slide

- Key definitions
- 2–3 cited findings from the literature

Gap + Research Question Slide

- The gap your study addresses
- Your research question

Scientific Storytelling: Literature Review Slide Practice

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- The gap your study addresses
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Practice presenting these two slides with a partner.

10:00